

Project SAJILO: PRD & Architecture

A Multi-Tenant, Multi-Category E-commerce Ecosystem for the Nepal Market

1. Executive Summary

SAJILO (meaning "Easy") is a localized platform designed to empower Nepali vendors to launch a full-featured, SEO-optimized e-commerce website instantly without technical overhead. The platform acts as a decentralized collection of vendor storefronts while aggregating all products into a centralized marketplace.

2. Target URL Structure

- **sajilo.com**: Central Marketplace (Next.js - SEO)
- **sajilo.com/foods**: Category Hub (Next.js - SEO)
- **sajilo.com/vendor-name**: Vendor Dynamic CMS Site (Next.js - High SEO)
- **sajilo.com/admin**: Vendor/Admin Dashboard (React/Vite - High Speed)

3. Technical Architecture

A. Infrastructure & Routing (The "No-Cost" Stack)

We avoid paid third-party modules by utilizing industry-standard open-source tools:

Layer	Technology	Role
Gateway	Nginx	Reverse proxy for routing traffic between Next.js (Storefront) and React (Dashboard).
Database	PostgreSQL	Shared instance using Schemas or RLS for multi-tenant isolation.
Microservices	Node.js / Fastify	Category-specific logic (Food, Electronics, Services).
Message Bus	Redis	Inter-service communication and caching.

4. The Agentic AI Skeleton Logic

The AI Agent's role is to automate the scaffolding of new microservices and dynamic vendor components. Below is the pseudo-code for the **Agentic Architect**:

```
class AgenticArchitect:
    def scaffold_service(self, category_name):
        # 1. Generate Database Schema for PostgreSQL
        schema = self.generate_schema(category_name)

        # 2. Scaffold Microservice Boilerplate (NestJS/Express)
        self.create_backend_module(category_name, schema)

        # 3. Create Next.js Dynamic Routes for SEO
        self.generate_frontend_routes(category_name)

        # 4. Inject into Nginx Config
        self.update_gateway_routing(category_name)

    def generate_vendor_cms(self, vendor_id, theme_preferences):
        # 1. Select UI Components (GSAP Animations)
        components = self.select_ui_library(theme_preferences)

        # 2. Build Next.js ISR Page
        return self.render_static_page(vendor_id, components)
```

5. UI/UX Design Philosophy

Frontend (Next.js): The "Bazaar" Experience

- **GSAP Product Rotations:** High-end animations for premium items (Woods/ Furnitures).
- **Localized Search:** AI-driven search that understands colloquial Nepali terms.
- **Mobile-First:** Optimized for 4G/LTE speeds common in Nepal.

Backend (React): The "Command Center"

- **Real-time Analytics:** Fast data fetching using TanStack Query.
- **Inventory Hot-loading:** Drag-and-drop interfaces for quick stock updates.

6. Database Strategy: Multi-Tenant PostgreSQL

To ensure security without multiple DB instances, we implement **Row Level Security (RLS)**:

```
-- Enable RLS
ALTER TABLE products ENABLE ROW LEVEL SECURITY;

-- Policy: Vendor can only see their own products
CREATE POLICY vendor_isolation_policy ON products
FOR ALL
USING (vendor_id = current_setting('app.current_vendor_id'));
```